



Assessment and Demonstration

Pilot- and Demonstration (P&D) Projects

Pilot- and demonstration projects test the technology and market feasibility. Their exposure raises public awareness and interest in the profession and paves the way for widespread adoption of innovation. Hence we initiated, built, or participated in, several P&D projects to test and demonstrate our coloured PV modules in collaboration with many industry partners. In 2016, we took part in the travelling road show of the national Energy Challenge, providing a PV box charging an electric racecar. In 2017 we built the PV façade “Swissness” at Umweltarena Schweiz, and in 2018 we were instrumental in realizing the innovative PV façade at the Solaris House and built the PV railing at NEST@empa.

Key words: Energy Challenge, Umweltarena Schweiz, Haus Solaris, NEST@empa

Target audience: building professionals, general public, PV manufacturing industry

2016 Energy Challenge

The Energy Challenge is a bi-annual national event to raise public awareness of energy efficiency and renewables through exhibition and experiences, travelling across many cities in Switzerland. We designed and supplied a PV glass box, featuring our coloured PV modules designed as Swiss cantonal flags and serving as shelter and charger of the world's fastest Formula Student e-racing car, built in collaboration between ETH and HSLU. The PV glass box was designed as off-grid PV system with battery storage and remote energy management system. The electrical system worked fault-free over the course of six months of travelling through eight cities, despite some minor water leakage through the roof. The NRP70 website featured this P&D project in their website's news section [1].



The HSLU glass box was graced by Swiss Solar Pioneer Bertrand Piccard and Stefan Nowak, national program leader PV research during the 15. National PV Conference at the Swisstec Convention Centre, Lausanne.

2017 PV façade “Swissness” at Umweltarena Schweiz

The Umweltarena is Switzerland's national public exhibition centre for energy efficiency and renewables in application, attracting more than 100'000 visitors annually. We initiated a collaboration with Umweltarena and Technology Transfer company ÜserHuus, and designed and built a PV façade on the

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adjacent staircase tower surrounding the entrance. The PV façade features all cantonal flags and the national flag as PV modules and hence underlines Umweltarena's claim of being a point of national interest. Since its inauguration in June 2017, all systems work properly, validating the technology's readiness [2]. The electrical and visual performance of the PV façade is monitored remotely and summarised live on a public display in the visitor centre.



The images show the original staircase tower (left) retrofitted with PV façade Swissness (middle) and collaborators Walter Schmid (Umweltarena, 2nd from left) and Jacqueline Schindler (ÜserHuus) during the inauguration on 22.6.2017.

Solaris House in Zürich

Architect Adrian Berger commissioned an Austrian PV manufacturer to build a terracotta coloured PV façade with vertically structured glass surfaces creating reflections resembling the dynamic water surfaces of the nearby lake. HSLU consulted on its meta-c-print specifications for digital ceramic print, which increased the electrical efficiency by 30% for the given terracotta colour [3].



Solaris House on the left and PV railing at NEST@empa on the right

PV railing at NEST@empa

This project features a new generation of our coloured PV modules, this time with mono-crystalline PV cells, designed and commissioned by HSLU with support of ÜserHuus. Three different designs (shutter, ornament and curve) were printed on glass of different finishes and reflection properties (float, silk and satinato). Each PV module is monitored individually so that the visual and electrical impact caused by designs and glass types can be compared. Preliminary results indicate that the satinato glass finish provides the best colour stability (or rather the least interference of printed and reflected colours), with almost identical electrical performance. This project is jointly developed with the SCCER Future and Energy Efficient Buildings and Districts (FEEBD).

References

- [1] From sun to fuel tank – colourful solar panels and electric vehicle. http://www.nfp70.ch/en/News/Pages/04072016_news_nfp70_energy-challenge.aspx
- [2] Umweltarena – Photovoltaik-Fassade "Swissness" in der Umwelt Arena Schweiz. <http://www.hslu.ch/umweltarena>
- [3] Haus Solaris by Huggenbergerfries Architects. <http://www.hbf.ch/projekte/wohnbauten/wohnhaus-solaris-zuerich/>
- [4] empa – Farbige PV Brüstung am NEST Gebäude. <http://www.hslu.ch/nest-pv>